Leo (Yuanzhe) Zeng

412-888-6280 | github.com/leoyzeng
leoz2@andrew.cmu.edu | Personal Website

EDUCATION

Carnegie Mellon University **Expected December 2027**
MS in Electrical and Computer Engineering

University of Waterloo **September 2021 – April 2026**
BASc in Computer Engineering GPA: 3.9/4.0 (91%)
President's Scholarship of Distinction and President's Research Award
Dean's Honor

EXPERIENCE

Cerebras Systems – Machine Learning Engineering Intern **September – December 2025**

- Developed and executed release test suites to validate performance, correctness, and reliability across software and hardware releases on Cerebras's wafer-scale AI chip, the world's largest and fastest machine learning accelerator.
- Implemented a telemetry and metrics pipeline using InfluxDB, enabling data-driven performance analysis and runtime optimization, and providing insights that guided future engineering improvements.

AMD (Advanced Micro Devices) – Software Engineering Intern **January – April 2025**

- Developed and optimized Windows display driver features for the latest generation of discrete GPUs.
- Diagnosed and resolved complex issues including display corruption, performance degradation, flickering, crashes, BSODs, black screens, and freesync/timing issues using WinDbg, ETL trace analysis, kernel-level debugging, register analysis, and memory dumps.
- Contributed to open-source Linux kernel by implementing bug fixes and adding debugging tools.

Reblink – Software Engineering Intern **January – April 2024**

- Developer of ARBO, a next generation sci-fi themed PvP strategy game where players command a squad of heroes with unique abilities to outsmart opponents in battles.
- Created a comprehensive tool to record and log game states and available actions, data collected is crucial for training a machine learning algorithm that uses transformers and convolution neural networks to predict optimal future moves.
- Designed and executed over 400 test cases for various in-game mechanics, documenting scenarios, results, and creating reports. This initiative saved the company $30,000 by eliminating the need for external testing services.

Dematic – Software Engineering Intern **May – August 2023**

- Developed the Dematic Virtual Facility, an efficient simulation and emulation platform capable of running over 10,000 robotics components in real-time, enabling comprehensive testing and optimization of a wide range of warehouse robotics and automation systems.
- Designed and implemented a robust communication interface (DCI Edge) based on documentation, which involved creating more than 30 message classes and transitioning from a fixed-length message serialization to a flexible JSON-based message serialization system, enhancing communication efficiency and flexibility.

Plan Group – Software Engineering Intern **September – December 2022**

- Increased efficiency for electrical and mechanical engineers by developing 5 software tools to automate and quality check CAD design, used by 125 colleagues daily.
- Created a tool to automatically quality check and revise 20,000+ wire segments in a CAD model, resulting in a process that is 300 times faster compared to manual quality check.

Collaborative Approach – Full Stack Software Developer Intern **January – April 2022**

- Improved appearance and user experience of front-end interface for file management system used daily by clinic staff, by replacing PHP-based functionality with interactive AJAX functionality.

- Provided clinic managers with a centralized, secure, and private document repository for employee communications in clinics in 5 cities, by implementing a SQL-based document storage and an HTML/JS/AJAX interface for creating and managing documents.

RESEARCH EXPERIENCE

University of Waterloo – Undergraduate Research Assistant **September – December 2024**
- Validated adaptive path following controllers for a nonholonomic mobile manipulator using MATLAB, ROS, and Symbolic Math Toolbox in both simulation and hardware experiments.
- Deployed control algorithms on a Clearpath Husky with a Kinova Gen3 arm, integrating Vicon motion capture data and troubleshooting system-level issues across hardware, networking, and code.

University of Waterloo – Undergraduate Research Assistant **May – Aug 2025**
- Conducted research on RISC-V CPUs with vector extensions to accelerate highly parallelizable tasks such as small neural network inference, evaluating open-source architectures through simulation, synthesis, and FPGA prototyping.
- Debugged and documented the full hardware build and simulation workflow producing a detailed technical guide that supports experiments in open-source CPU benchmarking and verification.

PROJECTS

Modular Robotics – Capstone Project **2025**
- led a 5-person team in developing a modular robotics platform, designing 12 interchangeable modules including motors, grippers, sensors, power, hubs, and displays.
- Designed and iterated custom 3D-printed module housings and mechanical connectors using Fusion 360, enabling quick assembly and reconfiguration of robotic components.
- Developed a distributed module architecture with wired inter-module communication and Wi-Fi connection to a PC, enabling arbitrary module configurations and a 3-minute setup time.

Sorting Algorithm Visualizer – Android Application **2022**
- Self-taught android development concepts such as activity/fragment life cycle, layout, xml, intents, and Kotlin to understand app development and app interface.
- Animated 4 sorting algorithms (selection, bubble, insertion, quick sort) to provide programming students with a visualization of sorting algorithms in real time.

Personal Website **2021**
- Self-taught JavaScript, CSS, HTML, React, and Bootstrap to better understand web development and user interface.
- Designed a multi-page website with responsive elements to display projects and pictures.
- Implemented an interactive slideshow that users can flip through to display project portfolio.

Chef Boy – 3D action-adventure game where the player explores an alien world **2020**
- Designed 40+ game entity classes with object-oriented concepts such as inheritance, abstraction, polymorphism, and visualized class interactions with UML diagrams.
- Utilized Blender to create 15 models & textures, and JMonkey Engine to display 3D graphics & lighting.
- Implemented data structures such as lists and queues to store items and AI enemies as objects for more efficient organization of game objects.

Escape Room Game – Embedded System **2021**
- Designed and implemented an escape room puzzle using STM32 Nucleo microcontroller and breadboards.

TECHNICAL SKILLS

Programming Languages	Python, C++, C, C#, Java, JavaScript , PHP, HTML/CSS, RISC-V, Verilog, MATLAB
Frameworks	Pytorch, Numpy, ROS, Unity, React, Laravel, jQuery, REST API
Tools	MySQL, InfluxDB, Git, Jira, Jenkins, Docker, AWS, UML, Postman, Selenium
Systems	Linux, Windows, Android, Kernel Debugging, WinDbg, GPU View
Hardware	Arduino, STM32, Raspberry Pi, FPGA, Vivado, Verilator, Fusion 360, Oscilloscopes, Soldering